

On the border of research and olympiads: erdosproblems.com

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1 Intro

In 2023, Thomas Bloom launched the website <https://www.erdosproblems.com/>. This is a database with many of Erdős' problems. The latter coming from 143 different problem collections. Erdős (1913-1996) was one of the most passionate and productive mathematicians, with a focus mostly in number theory and combinatorics (but actually almost all directions of maths). A few of his problems are extremely hard and important problems. Others are not important and very doable. Since the problems are automatically open for a while, especially when solved by young people it is sometimes a news item. In contrast to olympiad problems, it nevertheless may take very long to write up a proof. Wouter van Doorn spent 14 years to finally resolve one of the problems. In this short bundle, we will consider some examples of these problems. Possibly, later someone of you also solves one of the open questions.

1.1 Intuition on primes

On olympiads, the following theorems are not used usually, but it can give intuition on behaviour (what you expect to be the answer) as well.

Theorem 1.1 (Prime Number Theorem (PNT)). *The prime counting formula satisfies*

$$\pi(x) = \frac{x}{\ln(x)} + O\left(\frac{x}{\ln^2(x)}\right).$$

As a consequence, the n^{th} prime $p_n \sim n \ln(n)$.

Theorem 1.2 (Mertens).

$$\prod_{p \leq x} (1 - 1/p) \sim \exp(-\gamma) \frac{1}{\ln(x)}$$

(where $\exp(\gamma) \sim 1.781072$).

2 Examples

2.1 Erdős overlooked something?!

Erdős posed roughly more problems as there are in all IMOSL's together. So there are some of his personal longlist which would not make the shortlist. The following is one example.

Problem 2.1 (EP # 488). Let A be a finite set and $B = \{n \geq 1 : a \nmid n \ \forall a \in A\}$. Is it true that, for every $m > n \geq \max(A)$

$$\frac{|B \cap [1, m]|}{m} < 2 \frac{|B \cap [1, n]|}{n}?$$

2.2 Solved using computer code

Some questions were unsolved for 50 years, but thanks to computer power resolved recently.

Question 2.2 (EP # 399). Is it true that there are no solutions to

$$n! = x^k \pm y^k$$

with $x, y, n \in \mathbb{N}$, $xy > 1$, and $k > 2$?

The conditions are put to forbid the trivial solutions. While the number of solutions to $n! = x^2 - 1$ is not known, $n! = x^2 - y^2$ has of course infinitely many. Barfield found a solution by computer search, where $n! = x^4 - 36^4$.

Problem 2.3. What are the pairs $(n, x) \in \mathbb{N}^2$ satisfying $n! = x^4 - 36^4$, for $x < 2025$?

No solution is known for the sum of two large powers.

Question 2.4 (EP # 939). Let $r \geq 2$. An r -powerful number n is one such that if $p \mid n$, then $p^r \mid n$. If $r \geq 4$, then can the sum of $r - 2$ coprime r -powerful numbers ever be itself r -powerful?

Answer 2.5 (Cambie). Examples for $r = 5, 7, 8$

$$\begin{aligned} 3^7 \cdot 61^5 &= 2^8 \cdot 3^{10} \cdot 5^7 + 2^{12} \cdot 23^6 + 11^5 \cdot 13^5 \\ 3^7 \cdot 7^9 &= 2^{17} + 2^{23} + 2^{16} \cdot 5^8 + 5^{15} + 2^9 \cdot 13^7 \\ 2^{10} \cdot 3^{24} &= 31^9 + 2^{17} \cdot 3^{16} + 2^{16} + 2^{32} \cdot 3^{10} + 37^8 \end{aligned}$$

The question remains if there are infinitely many solutions.

2.3 Some recent extensive result

Question 2.6 (EP # 391). Let $t(n)$ be maximal such that there is a representation

$$n! = a_1 a_2 \cdots a_n$$

with

$$t(n) = a_1 \leq a_2 \leq \cdots \leq a_n.$$

Obtain good bounds for $\frac{t(n)}{n}$. In particular, is it true that

$$\lim_{n \rightarrow \infty} \frac{t(n)}{n} = \frac{1}{e}?$$

Problem 2.7. Do you see why the upper bound is true?

Answer 2.8 (Terence Tao).

$$\frac{1}{e} - O\left(\frac{1}{\log n}\right) \leq \frac{t(n)}{n} \leq \frac{1}{e} - \frac{c_0}{\log n} + o\left(\frac{1}{\log n}\right)$$

where $c_0 = 0.3044 \dots$ is an explicit constant (computed by an integral).

Tao conjectures the upper bound is sharp.

2.4 Some problems

Problem 2.9. Give an estimate for $\sum_{p \leq x} \frac{1}{p}$.

Problem 2.10. What is the proportion of positive integers that is squarefree?

Problem 2.11 (based on EP #689). For n sufficiently large, is there some choice of congruence class $c_p \pmod{p}$ for all primes $2 \leq p \leq n$ such that every integer in $[1, n]$ satisfies at least one congruences $\equiv c_p \pmod{p}$?

What if you want to have two congruences to be valid? (for n sufficiently large)

2.5 Some derived Olympiad questions

Problem 2.12 (RMM 2020). For each integer $n \geq 2$, let $F(n)$ denote the greatest prime factor of n . A strange pair is a pair of distinct primes p and q such that there is no integer $n \geq 2$ for which $F(n)F(n+1) = pq$. Prove that there exist infinitely many strange pairs.

This olympiad problem answers the following problem.

Question 2.13 (EP #649). Let $P(m)$ denote the greatest prime factor of m . Is it true that, for any two primes p, q , there exists some integer n such that $P(n) = p$ and $P(n+1) = q$?

Problem 2.14 (BxMO 2011). An ordered pair of integers (m, n) with $1 < m < n$ is said to be a Benelux couple if the following two conditions hold: m has the same prime divisors as n , and $m + 1$ has the same prime divisors as $n + 1$. (a) Find three Benelux couples (m, n) with $m \leq 14$. (b) Prove that there are infinitely many Benelux couples.

The pair $(m, n) = (75, 1215)$ is the only other example known (bounded by 2^{40}). This BxMO problem is related to the following problem by Erdős, for which of course no solution is known (but no proof of non-existence exists either).

Question 2.15 (EP # 850). There are examples where $\text{rad}(m) = \text{rad}(n)$, $\text{rad}(m+1) = \text{rad}(n+1)$ and $\text{rad}(m+2) = \text{rad}(n+2)$?

Here $\text{rad}(x)$ is the product of all its prime factors with multiplicity 1. E.g. $\text{rad}(2025) = 15$ (3, 5 being the only prime factors).

Also the following problem was related to one posed by Erdős (and Hall).

Problem 2.16 (Euler Olympiad 2025). We say that two positive integers m and n interlock if between every two divisors of n (both larger than 1) there exists a divisor of m , and between every two divisors of m (both larger than 1) there exists a divisor of n .

For example, $(63, 64)$ is an interlocking pair, as we can order their divisors as follows:

$$1 = 1 < 2 < 3 < 4 < 7 < 8 < 9 < 16 < 21 < 32 < 63 < 64,$$

and note that the divisors of 63 and 64 alternate here.

For which $k \in \mathbb{N}$, there exists two interlocking integers whose product equals the product of the first k primes?

We end with a problem, which is a special case of the Erdos-Selfridge theorem (that generalizes this to any product of $k \geq 2$ consecutive integers).

Problem 2.17 (Bosnia Herzegovina TST 2002). Prove that the product of $k = 10$ consecutive positive integers is not a perfect square.

3 Solutions

Problem 3.1 (EP # 488). Let A be a finite set and $B = \{n \geq 1 : a \nmid n \forall a \in A\}$. Is it true that, for every $m > n \geq \max(A)$

$$\frac{|B \cap [1, m]|}{m} < 2 \frac{|B \cap [1, n]|}{n}?$$

Answer 3.2. Let A be the set of primes in $[1, n]$, and $m = 2n$. Then the densities we are comparing are $\frac{\pi(2n) - \pi(n)}{2n}$ and $\frac{1}{n}$, the latter ratio being unbounded (and thus larger than 2).

Problem 3.3. What are the pairs $(n, x) \in \mathbb{N}^2$ satisfying $n! = x^4 - 36^4$, for $x < 2025$?

Answer 3.4. Since $37^4 - 36^4 > 36^3 > 7!$, $n! = x^4 - 36^4$ implies $12 \mid x$. If $36 \nmid x$, then $v_3(n!) = 4$, so $n < 12$. We obtain $10! = 48^4 - 36^4$, and for $x > 48$, since $5, 7 \nmid x$, we would have $x \geq 96$ and thus $x^4 - 36^4 > 2^4 \cdot 10! > 11!$.

If $36 \mid x$, so $x = 36z$, we have $n! = 36^4(z^4 - 1)$, where $n \geq 11$.

We have $z < 77$ since $x < 2025$. Since $7, 11 \nmid z^2 + 1$, $7, 11 \mid z^2 - 1 = (z-1)(z+1)$.

The latter results in two options modulo 77, $z = 34, 43$. Since then $n \geq 12$, we

need z to be odd, i.e. $z = 43$ is the other option. Then $z^2 + 1 = 1850 = 5^2 \cdot 37 \cdot 2$.

Checking v_5 and v_{37} , we get a contradiction.

The pair $(48, 10)$ is thus unique.

Problem 3.5. Do you see why the upper bound is true?

Answer 3.6. We have $\sqrt[n]{n!} \sim \frac{n}{e}$, and AM-GM.

Problem 3.7. Give an estimate for $\sum_{p \leq x} \frac{1}{p}$.

Answer 3.8. Using $\pi(k) = \frac{k}{\log k} \left(1 + O\left(\frac{1}{\log k}\right)\right)$, we have

$$\sum_{p \leq n} \frac{1}{p} = \sum_{k=1}^n (\pi(k) - \pi(k-1)) \cdot \frac{1}{k} = \sum_{k=1}^n \frac{\pi(k)}{k} - \sum_{k=0}^{n-1} \frac{\pi(k)}{k+1}.$$

This simplifies to

$$\begin{aligned} \frac{\pi(n)}{n} + \sum_{k=1}^{n-1} \pi(k) \left(\frac{1}{k} - \frac{1}{k+1} \right) &= \frac{\pi(n)}{n} + \sum_{k=1}^{n-1} \frac{\pi(k)}{k(k+1)} \\ &= \frac{\pi(n)}{n} + \sum_{k=1}^{n-1} \frac{\pi(k)}{k^2} - \sum_{k=1}^{n-1} \frac{\pi(k)}{k^2(k+1)} = \sum_{k=3}^{n-1} \frac{\pi(k)}{k^2} + O(1). \end{aligned}$$

Using the approximation for $\pi(k)$, we get

$$\sum_{k=3}^{n-1} \left[\frac{1}{k \log k} + O\left(\frac{1}{k \log^2 k}\right) \right] + O(1) = \log(\log n) + O(1).$$

Problem 3.9. What is the proportion of positive integers that is squarefree?

Answer 3.10. The answer is $\prod_p \left(1 - \frac{1}{p^2}\right)$.

Since $\frac{1}{\text{answer}} = \sum_{n \geq 1} \frac{1}{n^2} = \zeta(2) = \frac{\pi^2}{6}$, the answer is $\frac{6}{\pi^2}$.

Problem 3.11 (#689). For n sufficiently large, is there some choice of congruence class $c_p \pmod{p}$ for all primes $2 \leq p \leq n$ such that every integer in $[1, n]$ satisfies at least one congruences $\equiv c_p \pmod{p}$?

Answer 3.12. Iteratively, for every $2 \leq q \leq n$, let I_q be the integers that are not $\equiv c_p \pmod{p}$ for any $p < q$ and let c_q be the residue modulo q that appears the most in I_q . Then by Mertens theorem, there are less than $\frac{n}{2 \ln(n)}$ in I_p where p is the largest value smaller than $\frac{n}{10}$. Since we have $\pi(n) - \pi(0.1n) > \frac{n}{2 \ln(n)}$ many primes remaining, $|I_{p_{\pi(n)}}| = 0$.

Problem 3.13 (RMM 2020). For each integer $n \geq 2$, let $F(n)$ denote the greatest prime factor of n . A strange pair is a pair of distinct primes p and q such that there is no integer $n \geq 2$ for which $F(n)F(n+1) = pq$. Prove that there exist infinitely many strange pairs.

Solution 3.14. First, we prove the following lemma.

For primes $3 < q < r$, if $\text{ord}_q(2) = \text{ord}_r(2)$ then $(2, q)$ is a strange pair.

There must be a power of two among $n, n+1$, so n is either of the form $2^k - 1$ or 2^k . Therefore it suffices to show whenever $q \mid 2^k \pm 1$ that $F(2^k \pm 1) \neq q$.

Let d be the common order. If $q \mid 2^k - 1$, observe from $2^d - 1 \mid 2^k - 1$ that $F(2^k - 1) \geq F(2^d - 1) \geq r > q$. If $q \mid 2^k + 1$, observe from $2^d - 1 \mid 2^{2k} - 1$ and $q \nmid 2^k - 1$ that $2 \mid d$ and $F(2^k + 1) \geq F(2^{d/2} + 1) \geq r > q$.

For a large prime p , we have

$$2^{2p} + 1 = \left(2^p - 2^{(p+1)/2} + 1\right) \left(2^p + 2^{(p+1)/2} + 1\right).$$

These are relative prime factors, and except from one factor 5, the prime divisors of $2^{2p} + 1$ are larger. Hence there are $r > q > 5$ with other $(\text{ord}_q(2) = \text{ord}_r(2) = 4p)$ and we conclude.

Problem 3.15 (BxMO 2011). An ordered pair of integers (m, n) with $1 < m < n$ is said to be a Benelux couple if the following two conditions hold: m has the same prime divisors as n , and $m+1$ has the same prime divisors as $n+1$. (a) Find three Benelux couples (m, n) with $m \leq 14$. (b) Prove that there are infinitely many Benelux couples.

Answer 3.16. It is sufficient to prove that for $k \geq 2$, $(m, n) = (2^k - 2, 2^k(2^k - 2))$ is a Benelux couple. Clearly m and n have the same prime factors. Since $n+1 = 2^k(2^k - 2) + 1 = (2^k - 1)^2 = (m+1)^2$, the same is true for $m+1$ and $n+1$.

Problem 3.17 (Euler Olympiad 2025). We say that two positive integers m and n interlock if between every two divisors of n (both larger than 1) there

exists a divisor of m , and between every two divisors of m (both larger than 1) there exists a divisor of n .

For example, $(63, 64)$ is an interlocking pair, as we can order their divisors as follows:

$$1 = 1 < 2 < 3 < 4 < 7 < 8 < 9 < 16 < 21 < 32 < 63 < 64,$$

and note that the divisors of 63 and 64 alternate here.

For which $k \in \mathbb{N}$, there exists two interlocking integers whose product equals the product of the first k primes?

Answer 3.18. The answer is exactly every $k \in \{1, 2, 4, 6, 8\}$.

Here $k = 1$ is a silly case, attained by $n = 1$ and $m = 2$.

For $k \in \{2, 4, 6, 8\}$, take m and n to be the product of the first $\frac{k}{2}$ primes in $\{2, 5, 13, 19\}$ and $\{3, 7, 11, 17\}$ respectively.

When k is odd and at least 3, since the difference of number of prime factors between integer m and n with product the first k primes, needs to be at least $2^{\frac{k-1}{2}} \geq 2$, which cannot happen for two integers that interlock.

Finally, let k be larger than 8 and assume without loss of generality that m is divisible by 2. This implies $3 \mid n$ and $5 \mid m$, which (due to the divisor 10 of m) in turn implies that 7 and 11 must divide n . Continuing this process gives $p \mid m$ for $p \in \{2, 5, 13, 19, 23\}$ and $q \mid n$ for $q \in \{3, 7, 11, 17\}$. However, we now reach a contradiction as 23 and 26 are both divisors of m , while n cannot be a multiple of 24 or 25.

Problem 3.19 (Bosnia Herzegovina NMO 2002). Prove that the product of $k = 10$ consecutive positive integers is not a perfect square.

Answer 3.20. Note that the gcd of the integers is bounded by 10 and so no prime from 11 onwards can appear with odd exponent. Also, if $n(n+1) \dots (n+9)$ is a square, none of its factors is such a prime and thus n is quite large ($n > 113$ actually). It is easy to see that there is at most one perfect square among those. Among the 5 odd numbers, at most one is a multiple of 5, at most one a multiple of 7, and at most 2 a multiple of 3. Since the latter two are not both three times an odd square, we conclude.